Zhiyi Chen

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Education

Georgia Institute of Technology **Atlanta, GA**
Ph.D. Candidate in Computer Science; Advisor: Alberto Dainotti *Jan 2021 - Dec 2026 (expected)*
Thesis: Leveraging ML and AI for a multi-dimensional understanding of Internet Autonomous Systems
Master of Science in Computer Science; Concentration: Machine Learning *Jan 2021 - May 2025*
GPA: 4.0/4.0

Tsinghua University **Beijing, China**
Bachelor of Science in Computer Science and Technology *Aug 2016 - Jul 2020*
GPA: 3.73/4.0

Work Experience

ByteDance, Inc. **San Jose, CA**
Research Intern; Advisor: Peng Li *May 2026 - Aug 2026*

- Leading the development of a production-oriented Codex-based data-agent runtime for ByteDance's multimodal database, extending Codex with database-aware planning, skill routing, schema exploration/filtering, multimodal data exploration, and semantic-operator reasoning.
- Designed reliability and safety layers for sandboxed execution, SQL read/write policy, failure classification, bounded retry, audit tracing, and modular sub-agent orchestration for SQL critique and result verification.

Google LLC **Remote**
Student Researcher; Advisors: Alex Krentsel, Sylvia Ratnasamy *Jan 2026 - May 2026*

- Served as a main contributor to an internal multi-agent risk-assessment system for planned network changes, built with Google Agent Development Kit (ADK).
- Designed multi-agent workflows to jointly analyze heterogeneous internal documents, reports, and network telemetry for historical outage analysis and planned-change risk assessment.

ByteDance, Inc. **San Jose, CA**
Research Intern; Advisors: Peng Li, Tieying Zhang *May 2025 - Aug 2025*

- Developed TAHOE, a closed-loop execution-feedback learning framework for Text-to-SQL agents that implements an agent data flywheel by converting SQL execution traces into reusable experience through reflection, hint generation, retrieval, and attribution-based hint-bank management.
- Improved execution pass rate on the Spider2.0-Snow benchmark from 35.4% to 83.0% (+47.6 pp), reduced SQL debugging iterations by 11×, and demonstrated transferability across multiple LLM backbones.

Catchpoint Systems, Inc. **Remote**
CTG Intern; Advisor: Gavin Lynch *Jun 2024 - Aug 2024*

- Analyzed global Internet test data from measurement nodes targeting cloud and content providers.
- Designed automatic approaches for detecting false positives in inferred cloud outages and developed methods for detecting cloud performance degradations.

Coalition, Inc. **Remote**
Research Intern; Advisors: Daniel Woods, Morgan Hervé-Mignucci *Jun 2023 - Aug 2023*

- Leveraged public Internet measurement platforms and GIS tools to infer and visualize geographic locations of data centers that host policyholder domains.
- Explored the aggregation of cyber risks on physical infrastructure, especially critical points such as datacenters.

Publications & Preprints

- **Chen, Z.,** Song, J., Li, P. *"TAHOE: Text-to-SQL with Automated Hint Optimization from Experience."* arXiv preprint arXiv:2606.12387, 2026. Under submission to ACM SIGMOD 2027. (LLM Agents for Databases)
- **Chen, Z.,** Bischof, Z. S., Testart, C., Dainotti, A. *"AS2Biz: Leveraging Web Presence and AI to Improve AS Business Classification",* accepted at Internet Measurement Conference (IMC) 2026.
- **Chen, Z.,** Bischof, Z. S., Testart, C., Dainotti, A. *"Rethinking and Facilitating How We Classify Autonomous Systems by Network Properties",* accepted at Internet Measurement Conference (IMC) 2026.

- **Chen, Z.,** et al. *"AS2Org-Resolve: A Multi-Agent AI Pipeline for AS-to-Organization Mapping".* Under review at ACM Internet Measurement Conference (IMC) 2026.
- **Chen, Z.,** Bischof, Z. S., Testart, C., Dainotti, A. *"Improving the Inference of Sibling Autonomous Systems"*, published in the Passive and Active Measurement Conference (PAM) 2023.
- Li, P., **Chen, Z.**, Chu, X., Rong, K. *"DiffPrep: Differentiable Data Preprocessing Pipeline Search for Learning over Tabular Data"*, published at ACM SIGMOD 2023. (ML for data cleaning)
- **Chen, Z.,** Lin, T. **"***Principal Gradient Direction and Confidence Reservoir Sampling for Continual Learning",* published in the International Conference on Artificial Neural Networks (ICANN) 2021.

Invited Talks

- *"Improving the Inference of Sibling Autonomous Systems",* North American Network Operators' Group (NANOG) 87, Feb 2023, Atlanta.

Fellowships & Awards

- **Philip and Dianne Enslow Fellowship***,* College of Computing, Georgia Institute of Technology, 2026
- **Best Community Artifact Award**, Passive and Active Measurement Conference (PAM), 2023, for the artifact accompanying *"Improving the Inference of Sibling Autonomous Systems"*.